

Trig from complex

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this explains so much about angle

Aster after finding out that Math Prize for Girls is 90% trig

1 Theory

Complex numbers are vectors.

Example 1.1 (HMMT 2018/A6). If

$$\cos \alpha + \cos \beta + \cos \gamma = \sin \alpha + \sin \beta + \sin \gamma = 1,$$

compute the smallest possible value of $\cos \alpha$.

Example 1.2 (AIME I 2025/8). Let k be a real number such that the system

$$\begin{aligned} |25 + 20i - z| &= 5 \\ |z - 4 - k| &= |z - 3i - k| \end{aligned}$$

has exactly one complex solution z . Compute the sum of all possible values of k .

The great thing about complex numbers is that multiplication is angle addition, so they're better than normal vectors.

Example 1.3 (Machin's formula). $\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$.

Example 1.4. Show that

$$\tan(89.5^\circ) = \sin 1^\circ + \sin 2^\circ + \cdots + \sin 179^\circ.$$

Example 1.5 (CMIMC 2018/A7). Compute

$$\sum_{k=0}^{2017} \frac{5 + \cos(\frac{\pi k}{1009})}{26 + 10 \cos(\frac{\pi k}{1009})}.$$

2 Extras

If we have extra time, we can do this stuff I guess.

Lemma 2.1 (EGMO 6.28). *Let ABC be a triangle. It is equilateral if and only if $a^2 + b^2 + c^2 = ab + bc + ca$.*

Example 2.2 (CMIMC 2016/A6). For some complex number ω with $|\omega| = 2016$, there is some real $\lambda > 1$ such that ω , ω^2 , and $\lambda\omega$ form an equilateral triangle in the complex plane. Find λ .

Now let's have some fun.

Definition 2.3. A *cline* is a circle or a line.

This definition kind of makes sense if you view a line as a circle with infinite radius and center at infinity.

Definition 2.4. A function f is called a *Möbius transformation* if it is of the form

$$f(z) = \frac{az + b}{cz + d},$$

and f is nonconstant (or $ad - bc \neq 0$).

Note that any Möbius transformation is bijective. Now what do these two definitions have to do with each other? Well, we have the following nuclear weapon.

Theorem 2.5. *Möbius transformations take clines to clines. In other words, if f is a Möbius transformation and C is a cline, then the image of C under f is a cline.*

In algebraic geometry, the Möbius transformations are “morphisms of $\mathbb{P}^1$.” Don't worry, I don't know what that means either. We have another fun fact.

Theorem 2.6. *Given complex numbers A, B, C, X, Y , and Z such that A, B , and C are pairwise distinct and X, Y , and Z are pairwise distinct, there's exactly one Möbius transformation sending A to X , B to Y , and C to Z .*

This is rarely applicable, but when it is, it's so, so powerful.

Example 2.7 (Prize 2023/13). Let

$$f(t) = \frac{(10 + 9i)t - 10 + 9i}{t + i},$$

where $i = \sqrt{-1}$. Let $P = f(0)$, $Q = f(2023)$, and $R = f(1)$. Determine $\sin^2(m\angle PQR)$.

Example 2.8 (HMMT 2022/T6). Let $P(x) = x^4 + ax^3 + bx^2 + x$ be a polynomial with four distinct roots that lie on a circle in the complex plane. Prove that $ab \neq 9$.

Out of 99 teams and around 600 competitors, only *one* team solved the above problem in-contest, making it the hardest problem on the team round that year, despite its placement.

3 Problems

Problem 3.1. Derive the tangent addition identity $\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)}$ with complex numbers.

Problem 3.2 (Prize 2025/8). Compute

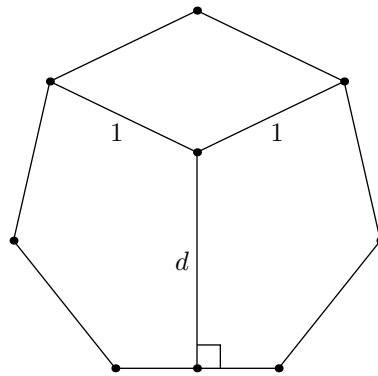
$$\sum_{n=0}^{\infty} \frac{\cos(\frac{\pi n}{6})}{\sqrt{3}^n}.$$

Problem 3.3 (Prize 2025/13). Compute A if $-90 \leq A \leq 90$ and

$$\sin(A^\circ) = \frac{\frac{1}{2} + \sum_{k=0}^{44} \sin(2k^\circ)}{\sum_{k=0}^{44} \sin((2k+1)^\circ)}.$$

Problem 3.4 (CMIMC 2019/8). The roots of the polynomial $P(z) = z^{2019} - 1$ can be written in the form $z_k = x_k + iy_k$ for $1 \leq k \leq 2019$. Let Q denote the monic polynomial with roots equal to $2x_k + iy_k$ for $1 \leq k \leq 2019$. Compute $Q(-2)$.

Problem 3.5 (Prize 2019/16). The figure shows a regular heptagon with sides of length 1.



Determine the indicated length d . Express your answer in simplified radical form.